

## **AN/SSQ-53F SONOBUOY CHARACTERISTICS**

The AN/SSQ-53F sonobuoy is a passive acoustic sensor used to detect, localize, and track a submerged submarine. Upon water entry the sonobuoy deploys a passive directional hydrophone (DIFAR) and a calibrated omnidirectional hydrophone (CO) to depths of 90, 200, 400, or 1000 feet. It also deploys an omni-directional hydrophone to 45 feet. This omni-directional hydrophone is called the Constant Shallow Omni (CSO) hydrophone. The sonobuoy is programmed prior to deployment through use of the electronic function select (EFS) circuitry accessed through the side of the SLC. The SSQ-53F incorporates Command Function Select (CFS) capability. This capability allows the operator on CFS-capable aircraft to: (1) toggle sonobuoy VHF transmitter power on and off, (2) change the RF channel, (3) turn automatic gain control (AGC) on (fast or slow) and off, (4) switch between the DIFAR hydrophone package, the CSO hydrophone, or the CO hydrophone, and/or (5) change the buoy life selection after buoy drop. The EFS is used to program the buoy prior to drop, and the CFS is used to change the programming after deployment.

| <b>TABLE 53F-2: AN/SSQ-53F CFS COMMANDS</b> |                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>COMMANDS</b>                             | <b>SELECTIONS</b>                                                                               |
| RF POWER                                    | ON (1.0 Watt), OFF                                                                              |
| RF CHANNEL                                  | 1 – 99, except 57, 58, & 93 (Note: These channels will not appear as options.)                  |
| ACOUSTIC GAIN / AGC                         | AGC OFF, ON SLOW (A1), ON FAST (A2)<br>(Note: Defaults to SLOW regardless of selection when ON) |
| MODULATION INPUT                            | DIFAR, CSO, or CO                                                                               |
| LIFE                                        | 0.5, 1, 2, 4, or 8 hours                                                                        |

## **AUTOMATIC GAIN CONTROL (AGC)**

AGC is available in DIFAR mode only; it is not available in the CSO hydrophone. The AGC measures the acoustic energy in the DIFAR omnidirectional channel and uses this data to set the gain of all three DIFAR channels.

The AGC Fast selection is inoperative. Thus, if AGC Fast is selected via the EFS, the buoy will process using AGC Slow. In effect, the buoy only has an AGC ON (either AGC Fast or AGC Slow selected) or an AGC OFF selection. Initial production of the AN/SSQ-53F is similar to the AN/SSQ-53E (AGC Fast is inoperative). AGC Slow (Narrowband AGC) continuously samples ambient noise from the omni-directional channel and adjusts the DIFAR channels accordingly. If high ambient noise is present, the AGC circuitry will reduce the gain in the DIFAR channels to lessen the probability of transmitter saturation (i.e., reaching the maximum FM transmitter deviation level). Slow AGC was designed for use in areas of high sea state, heavy shipping activity, or rain (Sea State 3 – 6 on the

Beaufort Scale or Wenz Curves). Table 53F-3 provides some example ambient noise levels for various frequencies where Slow AGC might be advantageous.

### NOTE

Currently the AN/SSQ-53F sonobuoys are made with the Fast AGC selection inoperative. If Fast AGC is selected on these buoys, the buoy defaults to Slow AGC.

| TABLE 53F-3: Slow AGC Ambient Noise Levels |          |
|--------------------------------------------|----------|
| FREQUENCY                                  | AN LEVEL |
| 40 – 50 Hz                                 | 85-90 dB |
| 100 Hz                                     | 75-80 dB |
| 1000 Hz                                    | 60 dB    |

If low ambient noise levels are detected in the omni-directional channel the Slow AGC circuitry will increase the gain in the DIFAR channels. This will amplify weak signals above the noise floor allowing recognition by the operator. Slow AGC adjusts the gain of the DIFAR channels at approximately 0.7 dB per second linearly (about 0.5 dB per second in steps) until the output returns to normal.

If AGC is selected on (AGC Slow or AGC Fast) the acoustic signal will go to their AGC on values. When AGC is selected off the acoustic signal immediately returns to the non-AGC levels.

### **CALIBRATED OMNI HYDROPHONE**

All AN/SSQ-57B Calibrated LOFAR sonobuoys have expired. The capability for high frequency processing was incorporated in the AN/SSQ-53F with the development of the Calibrated Omni (CO) hydrophone. This hydrophone is selected for use prior to launch via the EFS or after launch via CFS. When the CO hydrophone is selected, any other hydrophone data, i.e., DIFAR or CSO, is not available to the operator.

The CO hydrophone is physically located approximately three feet above the DIFAR package on the sonobuoy compliance. Its frequency response curve is calibrated from 5Hz – 20 KHz, with minimal frequency response roll-off to 25 KHz. This means that usable data is available from 5Hz – 25 KHz but is only calibrated to 20 KHz. Accuracy of the signal from 5 Hz – 20 KHz, with the calibration curves provided for each unit, is  $\pm 1$  dB. The sonic frequency response diagram for the CO hydrophone on the AN/SSQ-53F is located in the frequency response section of this IETM.

## **AN/SSQ-53F ELECTRONIC FUNCTION SELECT**

The AN/SSQ-53F EFS circuitry, Figure EFS-2, allows the operator to program sonobuoy RF channel, operating life, depth, operating sensor, and AGC mode selection settings electronically instead of mechanically. The EFS will operate through at least 50 selection cycles (30 - 60 seconds long) per the AN/SSQ-53F specification.

### **AN/SSQ-53F EFS OPERATION**

On the side of each sonobuoy, visible through the sonobuoy launch container (SLC) via clear plastic 'windows' near the muzzle end, are the EFS buttons and readout, see Figure EFS-2. The digital readout is a light emitting diode (LED) located above the two buttons, and is made up of two seven-segment digital readouts. The buttons are positioned next to each other in the horizontal plane; the left button is the "SET" button and the right button is the "VERIFY" button. The SET button is depressed to cycle lists the EFS set cycle.

Upon depressing and releasing the SET button for the first time, EFS circuitry power is applied and the LED readout "Tens" digit (the left hand digit) begins to cycle from '0' to '9' in one second intervals. Upon reaching the desired number, the operator depresses and releases the SET button again, freezing the Tens digit at its current number and starting the "Ones" digit (right hand digit) cycling from '0' to '9' in one second intervals. Upon reaching the desired number in the "Ones" digit position, the operator presses and releases the SET button freezing the "Ones" digit at its current number. This concludes the RF channel selection.

After the RF channel has been selected, the life selection begins to cycle through the possible selections in one-second increments. Selections are 30 minutes, one, two, four or eight hours. The LED readouts will illuminate with '0.5' (30 minutes), '1.0' (one hour), '2.0' (two hours), '4.0' (four hours), or '8.0' (eight hours.) Depressing the SET button while the desired life is illuminated sets the life. This concludes the life selection operation.

One second after the life selection is set, the depth selection begins to cycle through 90 feet, 200 feet, 400 feet, or 1000 feet in one-second increments. The LED readout will illuminate with "d1" for 90 feet, "d2" for 200 feet, "d3" for 400 feet, and "d4" for 1000 feet. Depressing the SET button while the desired depth code is illuminated sets the depth. This concludes the depth selection operation.

The EFS numerals begin to cycle through the 'Operating Sensor' selections of constant shallow omni (CSO), DIFAR, or Calibrated Omni-directional LOFAR ('CS,' 'dF,' or 'CO'.) If the operator depresses the VERIFY button at this point, instead of depressing the SET button to select the 'Operating Sensor,' the buoy will revert to the operating sensor and AGC mode it was last set to plus the RF, life and depth selections just made. It is strongly recommended that the EFS be fully programmed rather than using the VERIFY button at this point in the programming sequence. If the operator depresses the SET button at this point in the programming sequence to specify what 'Operating Sensor' is used, that sensor is set, and the AGC mode must also be selected. This concludes the 'Operating Sensor' selection operation.

The EFS cycles through the AGC modes (AGC OFF, AGC ON SLOW, or AGC ON FAST) in one-second increments. The LED readout will illuminate with "OF" for AGC off, "A1" for AGC on

Slow and “A2” for AGC on Fast. Depressing the SET button while the desired AGC mode is illuminated sets the AGC mode. Note that in the AN/SSQ-53F the AGC defaults to Slow AGC even if Fast AGC is selected. Fast AGC will become functional in future contracts and versions of the SSQ-53. This concludes the AGC mode selection operation.

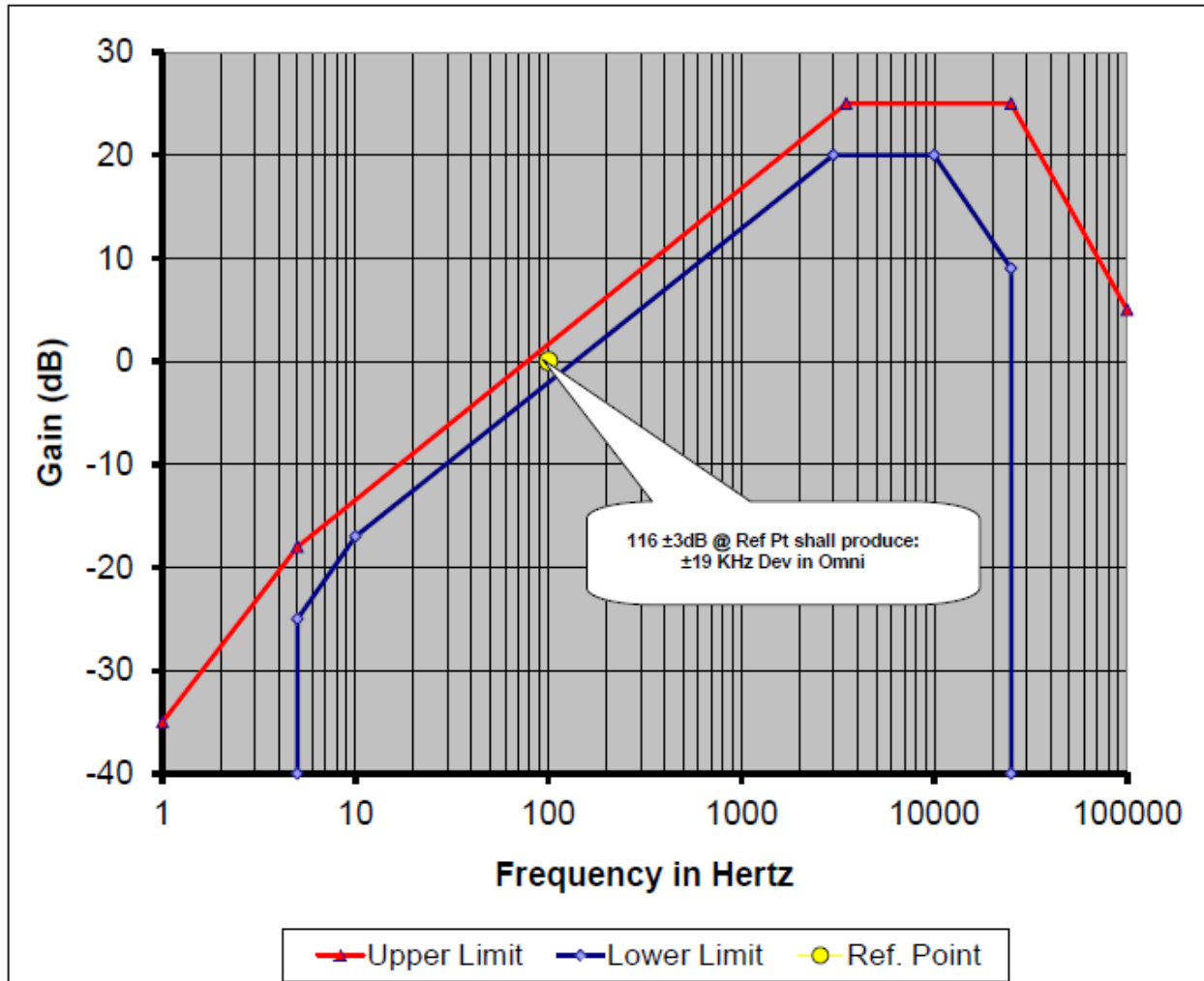
Setting the EFS normally takes the operator from 30 - 60 seconds. The RF channel number, life, depth, operating sensor, and AGC mode selections set in the EFS will be retained in sonobuoy memory for at least 48 hours per the specification. The VERIFY button is used to review and verify settings made in the EFS. Depressing the VERIFY button causes the contents of memory to be displayed on the EFS readout for three seconds and restarts the 48 hour retention period. Specifications call for the EFS circuitry to function for a minimum of 50 “Set” cycles and 100 “Verify” cycles before battery exhaustion.

| Table EFS-2: AN/SSQ-53F EFS ‘SET’ Button Function |                                                                                                                     |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| SET Switch Depression Number                      | Function                                                                                                            |
| 1 <sup>st</sup>                                   | Activates EFS, displays ‘0’ in ‘Tens’ digit, begins cycling of ‘Tens’ digit.                                        |
| 2 <sup>nd</sup>                                   | Sets ‘Tens’ digit with displayed RF channel number. Begins cycling of ‘Ones’ digit.                                 |
| 3 <sup>rd</sup>                                   | Sets ‘Ones’ digit with displayed RF channel number. Starts cycling of ‘Life’ selections (0.5, 1, 2, 4, or 8 hours). |
| 4 <sup>th</sup>                                   | Sets ‘Life’ selection. Starts cycling of ‘Depth’ selections (d1=90, d2=200, d3=400, or d4=1000 feet)                |
| 5 <sup>th</sup>                                   | Sets the depth selection. Begins cycling through ‘Operating Sensor’ (CS, dF or CO) selections.                      |
| 6 <sup>th</sup>                                   | Sets ‘Operating Sensor.’ Begins cycling through AGC setting (OF, A1 <sup>Slow</sup> , or A2 <sup>Fast</sup> )       |
| 7 <sup>th</sup>                                   | Sets AGC status. Loads RF, life, depth, operating sensor, and AGC mode selections into memory, turns off EFS power. |

## AN/SSQ-53F SONOBUOY SONIC FREQUENCY RESPONSE

The AN/SSQ-53F incorporates the 'constant shallow omni (CSO) omni-directional hydrophone deployed to a depth of 45 feet in addition to the regular DIFAR hydrophone. The SSQ-53F also incorporates the calibrated omni-directional (CO) hydrophone. The CO hydrophone is a calibrated LOFAR hydrophone that replaces the functionality of the SSQ-57 (Series) sonobuoy.

SSQ-53F CO Frequency Response



SSQ-53F CSO Frequency Response

